SSA4 (P22202) — Plasma Membrane (GO:0005886) Annotation Review

Focus: existing_go_annotation_decision · `existing_annotations[3]` in `genes/yeast/SSA4/SSA4-ai-review.yaml` **Seed hypothesis:** Keep the IBA plasma-membrane annotation as **KEEP_AS_NON_CORE**. **Gene:** SSA4 / *Saccharomyces cerevisiae* stress-inducible cytosolic Hsp70 (SSA subfamily), 642 aa.

Executive Judgment

Verdict: Over-annotated — the seed KEEP_AS_NON_CORE action is defensible but arguably too weak; REMOVE is the better-supported lead.

The plasma-membrane (GO:0005886) annotation on SSA4 is a **phylogenetic (IBA, GO_REF:0000033) inference with no yeast-specific support**. Three independent lines of evidence converge:

1. **No experimental support in yeast.** UniProt records SSA4 subcellular location as *Cytoplasm* only. The only experimental (IDA, P 11279056) cellular-component evidence is for cytoplasm and nucleus. SSA4 is a nucleocytoplasmic shuttling Hsp70 (P 17020589).
2. **The PM annotation is not present in the current GO release.** A live QuickGO query for P22202 returns six cellular-component annotations — nucleus, cytoplasm, cytosol (IDA/IEA/IBA) — and **no plasma-membrane annotation**. GO:0005886 survives only as a stale cross-reference in the cached UniProt entry. In other words, GO_Central/PAINT appears to have already dropped this term for the Hsp70 ancestral node relevant to Ssa4.
3. **The proximal source of the family PM signal is a single contamination-prone yeast PM-proteome study — which did not include Ssa4.** Live QuickGO shows PM (GO:0005886) annotations for the constitutive paralogs **Ssa1, Ssa2, and Ssb1**, all **HDA** evidence from **one**

reference,

16622836

(Delom et al. 2006, stripped-PM-fraction mass-spec of ~90 proteins). Abundant cytosolic chaperones routinely co-purify in such membrane fractions. **Ssa4 was not detected** (it is stress-inducible with low basal abundance), so its PM term was purely phylogenetic (IBA) and is now absent from live GO. (A separate mammalian tumor-cell-surface Hsp70 phenomenon exists — Gb3/lipid-raft dependent,

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— but human HSPA1A itself carries no PM GO term, so this is not the direct source here.)

4. **The authoritative organism database (SGD) does not annotate Ssa4 to the plasma membrane.** SGD's GO details for YER103W (146 annotations) list cellular-component terms only for **cytoplasm and nucleus** (manually curated) plus cytoplasm/nucleus/cytosol (computational) — **no GO:0005886**. SGD's curated description attributes Ssa4's membrane role to "**SRP-dependent cotranslational protein-membrane targeting and translocation**" (i.e., the **ER**) and describes it as a "cytoplasmic protein that concentrates in nuclei upon starvation."

Most important caveat: Cytosolic Hsp70s (including yeast Ssa) genuinely engage membranes peripherally during protein/mRNA targeting — but at the **ER** (SRP-dependent cotranslational translocation, per SGD) and the **mitochondrial outer membrane** (Tom70-dependent mRNA/precursor delivery,

22138184

), **not the plasma membrane**. The paralog HDA "PM" detections are a weak, contamination-prone basis, and even they do not include Ssa4. So the "plausible peripheral association" rationale in the seed is real but thin, and does not specifically implicate Ssa4 at GO:0005886.

Evidence Matrix

Citation	Evidence type	Supports/ Refutes	Claim tested	Key finding	Context	Co lim
UniProt P22202 (record)	Database	Refutes PM	Primary localization of Ssa4	Subcellular location = "Cytoplasm" only	*S. cerevisiae*	Hig cur sur dat lev
QuickGO live query (P22202)	Database/ computational	Refutes PM	Does GO currently annotate PM?	6 CC annotations (nucleus, cytoplasm, cytosol); **no GO:0005886**	GO_Central release	Hig cur ma PAI edi
P 11279056 (IDA, SGD)	Localization (experimental)	Qualifies	Experimental localization	IDA support only for cytoplasm and nucleus	*S. cerevisiae*	Hig nuc sile
P 17020589	Localization / transport	Refutes PM	Where does Ssa4 reside?	Ssa4p shuttles nucleus↔cytoplasm; nuclear on stress via Msn5 export	*S. cerevisiae*	Hig me loc
P 22138184	Mutant/ mechanism	Qualifies	Membrane association of Ssa	Ssa1 targets mRNA to **mitochondrial outer membrane** (Tom70)	*S. cerevisiae*	Per org not me
P 25853343	Interaction/ mechanism	Qualifies	Ssa membrane/ pore contacts	Ssa2 binds nucleoporin Nup116 for tRNA import	*S. cerevisiae*	Nu not me
P 16622836 (Delom 2006)	Localization (HDA proteomics)	Qualifies / competing origin	Are yeast Hsp70s at the PM?	Ssa1/Ssa2/Ssb1 detected in stripped-PM fraction (~90 proteins); **Ssa4 not detected**	*S. cerevisiae*	HD par onl abu cha pro pur no

Citation	Evidence type	Supports/ Refutes	Claim tested	Key finding	Context	Co lim
P 37932378	Structural/ biophysical	Competing (mammalian)	Basis of PM-Hsp70	Hsp70 binds DOPC/ DOPS bilayers; PM-Hsp70 specific to tumor cells	Mammalian/ artificial membrane	Dis phe hur HSI car GO
P 30632067; P 39015084	Localization (mammalian)	Competing (mammalian)	Where is surface Hsp70 seen?	mHsp70 on tumor-cell surface, Gb3/ lipid-raft dependent	Human/ mouse tumor	Org cor spe
QuickGO paralog scan (this run)	Computational/ database	Qualifies	Which Hsp70s carry PM in GO?	PM present for Ssa1/Ssa2/Ssb1 (HDA, P 16622836); absent for Ssa4, Ssa3, human HSPA1A	GO_Central live	Dir ref cur rel
SGD YER103W record (this run)	Review/ database	Refutes PM	Does SGD place Ssa4 at PM?	CC = cytoplasm/ nucleus/cytosol only; membrane role = SRP-dependent **ER** cotranslational targeting	*S. cerevisiae*	Aut org no

GO Curation Implications

- **Term:** plasma membrane (GO:0005886), a **cellular component (CC)** term. Evidence type IBA / GO_REF:0000033.
- **Recommended lead (requires curator verification): REMOVE** the plasma-membrane annotation, or at minimum retain as **NON_CORE** only if the curator prefers strict conservatism. Rationale: (a) no yeast experimental support; (b) the term is absent from the

current GO_Central set for P22202, suggesting the phylogenetic inference has already been retracted upstream; (c) the ancestral signal derives from mammalian tumor-cell-surface Hsp70, an organism/context-specific property.

- **Better-supported CC terms are already present and experimental:** cytoplasm (GO:0005737, IDA), nucleus (GO:0005634, IDA), cytosol (GO:0005829). These should be the core localizations.
- The seed action **KEEP_AS_NON_CORE is not wrong**, but it is **more generous than the evidence warrants**. The only yeast PM evidence for this family (HDA, P 16622836) covers the paralogs Ssa1/Ssa2/Ssb1, is contamination-prone, and **excludes Ssa4**; the genuine peripheral membrane roles of Ssa chaperones concern the **ER/mitochondrial outer membrane**, not the plasma membrane. If the curator prefers conservatism, KEEP_AS_NON_CORE is acceptable, but the annotation should be explicitly flagged as an IBA no longer present in the live GO release for Ssa4.

Mechanistic Scope

- **Direct molecular function of Ssa4:** ATP-dependent Hsp70 chaperone (protein folding, nascent-chain and stress-denatured protein binding), operating in the **cytosol and nucleus**.
- **Genuine membrane-proximal roles (peripheral, transient):** delivery of precursor proteins/mRNAs to the **ER (Sec translocon)** and **mitochondrial outer membrane (Tom70)**; interaction with **nuclear pore** components (Nup116, Nup82) during nucleocytoplasmic shuttling. None of these are the plasma membrane.
- **Not supported as direct gene-product localization:** residence in/at the plasma membrane.

Paralog / ortholog PM-annotation comparison (live QuickGO, this run)

Protein	Accession	PM (GO:0005886)?	Evidence	Reference
Ssa4 (yeast, target)	P22202	**No**	—	— (IBA only, now absent)
Ssa1 (yeast)	P10591	Yes	HDA	P 16622836
Ssa2 (yeast)	P10592	Yes	HDA	P 16622836
Ssa3 (yeast)	P09435	No	—	—
Ssb1 (yeast)	P11484	Yes	HDA	P 16622836
Ssc1 (yeast, mito)	P12398	No	—	—
HSPA1A (human)	P0DMV8	No	—	—
HSPA8 (human)	P11142	Yes	IEA/TAS	—

Conflicts and Alternatives

- **Real-but-weak yeast paralog evidence:** Unlike my initial mammalian-origin hypothesis, the yeast PM annotations for Ssa1/Ssa2/Ssb1 are experimental (HDA) — but all from **one** stripped-PM-fraction proteomics study (P 16622836) where abundant cytosolic chaperones commonly co-purify, and which **did not include Ssa4**. This is a contamination-prone, non-specific basis and is paralog-, not Ssa4-, specific.
- **Paralog/ortholog carry-over:** The Hsp70/HSPA PANTHER family mixes yeast Ssa, Ssb, Sse, mammalian HSPA1A/HSPA8, etc. The separate mammalian tumor-cell-surface Hsp70 biology belongs to inducible HSPA1A and should not be propagated to Ssa4; note HSPA1A itself carries no PM GO term.
- **Stale vs. live records:** UniProt's cached cross-reference still lists GO:0005886 (IBA), whereas live GO_Central/QuickGO does not — a database-lag artifact that could make the annotation look current when it is effectively retired.

- **Real-but-different membrane biology:** Ssa's authentic ER/mitochondrial-membrane peripheral associations could be mistaken as generic "membrane" support, but they map to distinct CC terms, not plasma membrane.

Knowledge Gaps

1. **Is the PM annotation truly retracted upstream, or filtered by QuickGO?** — Checked: live QuickGO returns no GO:0005886 for P22202. Matters because it changes REMOVE vs KEEP. Resolution: inspect the current PAINT/PANTHER family tree (PTN node) annotations and the GO_Central GAF directly.
2. **Any high-throughput yeast plasma-membrane/proteomics hit for Ssa4?** — Not found in targeted literature. Matters because a peripheral PM proteomics signal could justify NON_CORE. Resolution: check membrane-proteome and BioID/proximity datasets (SGD, GFP-localization Huh et al.).
3. **Does any yeast Ssa localize to the plasma membrane under stress?** — No evidence found; stress redistributes Ssa4 to the nucleus, not the PM. Resolution: stress-condition live imaging.

Discriminating Tests

- **GFP/mNeonGreen live imaging of Ssa4** under normal and heat/ethanol stress with a PM marker (e.g., Pma1) — expect cytosol/nucleus, no PM enrichment.
- **Cell-surface biotinylation / non-permeabilized immunostaining** for Ssa4 — expect negative (unlike mammalian tumor Hsp70).
- **Subcellular fractionation** (cytosol vs. PM vs. ER vs. mitochondria) with quantitative MS — expect cytosolic/nuclear, with minor ER/mito peripheral pools, not PM.
- **Inspect the current PANTHER PTN node GAF** to confirm whether GO:0005886 is still propagated to the Ssa4-containing subtree.

Curation Leads (require curator verification)

- **Action change lead:** Consider **REMOVE** for GO:0005886 (IBA) rather than KEEP_AS_NON_CORE; if retained, keep strictly as NON_CORE, low-confidence.
- **Candidate references / snippets to verify:**
- P 17020589
  — "*Cytoplasmic hsp70s like yeast Ssa4p shuttle between nucleus and cytoplasm under normal growth conditions but accumulate in nuclei upon stress.*" (supports cytosol/nucleus as the real localizations).
- P 37932378
  — "*Membrane-bound heat shock protein 70 (Hsp70) apart from its intracellular localization was shown to be specifically expressed on the plasma membrane surface of tumor but not normal cells.*" (identifies mammalian tumor origin of PM-Hsp70).
- P 11279056
  — IDA source for cytoplasm and nucleus (SGD).
- **Candidate retained CC terms:** GO:0005737 (cytoplasm, IDA), GO:0005634 (nucleus, IDA), GO:0005829 (cytosol) as core.
- **Suggested question for curator:** Is the reviewed PM IBA still present in the GO_Central GAF/PAINT tree, or is it already retired (as live QuickGO suggests)? If retired upstream, the review row should note the annotation no longer exists and mark REMOVE/obsolete rather than KEEP.
- **Suggested experiment:** cell-surface biotinylation + fractionation MS to definitively exclude a yeast plasma-membrane pool.

Provenance

- UniProt REST (`P22202.json`): subcellular location + GO cross-references (executed).
- QuickGO annotation API (`/annotation/search`, geneProductId=P22202): 20 total annotations, 6 CC, **no GO:0005886**; IBA `withFrom` ortholog list captured (executed).
- QuickGO paralog/ortholog scan (Ssa1/Ssa2/Ssa3/Ssb1/Ssc1, HSPA1A/HSPA8/mouse Hsp70): PM present only for Ssa1/Ssa2/Ssb1 (HDA, all

)

P 16622836

(executed).

- SGD backend API (`/backend/locus/YER103W` + `/go_details`): 146 GO annotations, CC = cytoplasm/nucleus/cytosol only, no PM; curated description cites SRP-dependent ER cotranslational targeting (executed).
- PubMed searches:

P 17020589

P 11279056

P 22138184

P 25853343

P 16622836

P 37932378

P 30632067

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P 39015084

All computational results above were executed against live public resources during this run; no results were fabricated. UniProt's cached GO cross-reference listing GO:0005886 (IBA) conflicts with the live QuickGO set, which is reported as a database-lag caveat rather than resolved definitively.
